

1 Application-relevant settings in HYDRA

Configuring variants for Stand-alone CAQ

The request for inspection specifications is based on a uniform selection table. Opposite to the integration scenarios with PP-PDC and/or HYINFO, for stand-alone CAQ integration it is not possible to use any data provided by SAP to request the specifications.

For the following reason there is a customizing table in HYDRA. The table represents the QAILS selection structure defined by SAP and allows though, requesting inspection specifications from SAP. Right now there is no graphical user interface to maintain the table, though any adjustments have to be done by using database interface.

Table: **CAQ_QAILS_VORMERK**

Field	T	L	Description	Meaning / Remark
SATZART	CHAR	3	Record type for request record	Fixed „Q40“
LOSNR_VON	NUMC	12	From inspection batch number	Should not be used for selection
LOSNR_BIS	NUMC	12	To inspection batch number	Should not be used for selection
PLNFL	CHAR	6	Operation sequence in task list	Should not be used for selection
VORNR_VON	CHAR	4	From operation number	Should not be used for selection
VORNR_BIS	CHAR	4	To operation number	Should not be used for selection
VORGWERK	CHAR	4	Plant of operation to be processed	Should be used for selection
SUBSYS	CHAR	6	Identifier of the subsystem	Value as defined in SAP customizing Default: QM0001
PRPLATZ	CHAR	8	Work center	Useable for selection
PRPLATZWRK	CHAR	4	Plant of the work center	Useable for selection
MATNR	CHAR	18	Material number	Useable for selection
DATUM_VON	CHAR	15	From creation date of inspection batch	It is possible to calculate the “date from” dynamically according to the current date minus x days. For that the entry has to be done such as: TODAY-n (n represents the number of days, e.g. 5) Default: TODAY
DATUM_BIS	CHAR	15	To creation date of inspection batch	It is possible to calculate the “date from” dynamically according to the current date minus x days. For that the entry has to be done such as: TODAY-n (n represents the number of days, e.g. 5) Default: Today
PRUEFSTAT]	CHAR	1	Status of the inspection	Useable for selection
ART	CHAR	8	Inspection type	Useable for selection
HERKUNFT	CHAR	2	Origin of the inspection batch	Useable for selection

Field	T	L	Description	Meaning / Remark
CHARG	CHAR	10	Batch number	Should not be used for selection
AUFNR_VON	CHAR	12	From order number	Should not be used for selection
AUFNR_BIS	CHAR	12	To order number	Should not be used for selection
LIFNR	CHAR	10	Vendor number	Should not be used for selection
KUNNR	CHAR	10	Customer number	Useable for selection
MBLNR	CHAR	10	Number of the material document	Should not be used for selection
MAXLOSANZ	NUMC	4	Maximum number of batches per transmission	Fixed "9999"
BEARB	CHAR		User	Future use
TA_ID	CHAR	30		Future Use
STATISCH	CHAR	1	Flag "Static entry"	Fixed "J"
STATUS	CHAR	3		Future use
BEARB_DATE	DATE			Future use
BEARB_TIME	TIME			Future use
USER_D_01	DATE		User field	Future use
USER_D_02	DATE		User field	Future use
USER_N_03	NUMC	8	User field	Future use
USER_N_04	NUMC	8	User field	Future use
USER_N_05	NUMC	8	User field	Future use
USER_F_07	DEC	13,3	User field	Future use
USER_F_08	DEC	13,3	User field	Future use
USER_C_09	CHAR	20	User field	Future use
USER_C_10	CHAR	40	User field	Future use
Verweis			Database serial	Consecutive number

Use the [HYDRA Scheduler](#) to maintain entries for the download of inspection specifications in CAQ stand-alone scenario. The program has only to be called once per subsystem. It will then execute all entries in the CAQ_QAILS_VORMERK table one after the other

Parameter name	Value
Product key	SAP-QMIDI
License key	SAP-QM-IDI
Command (Windows):	sh.exe .hysapqmc.scr /MESTYP_OUT=ZQM_IDI /VARIANTE=QM_IDI /LOGSYS=<created logical system>
Command (Unix):	./hysapqmc.scr /MESTYP_OUT=ZQM_IDI /VARIANTE=QM_IDI /LOGSYS=<created logical system>
Comment:	SAP-QMIDI: Download inspection specifications

Parameter name	Value
Intervall	5

Configuring call for catalog master data

Catalog master data of SAP is needed for all integration scenarios. The catalog master data download is also executed by the hysapqmc.exe/out program. Use the [HYDRA Scheduler](#) to maintain entries for the download of catalog master data.

Parameter name	Value
Product key	SAP-QMIDI
License key	SAP-QM-IDI
Command (Windows):	sh.exe ./hysapqmc.scr /MESTYP_OUT=ZHYQMIDI_CATALOG /LOGSYS=<created logical system> /VARIANTE=< variant as defined in table SAP_SF_PARAM_CFG >
Command (Unix):	./hysapqmc.scr /MESTYP_OUT=ZHYQMIDI_CATALOG /LOGSYS=<created logical system> /VARIANTE=< variant as defined in table SAP_SF_PARAM_CFG >
Comment:	SAP-QMIDI: Download catalog master data
Intervall	5

The HYDRA scope of delivery includes the “QM_IDI” variant for the QIRF_SEND_CATALOG_DATA2 function module as an example. In the example all parameter values are empty.

If necessary multiple entries have to be created to download all necessary catalog master data from SAP. It is strictly recommended to create the variants in the customer name space, starting with “U_”.

Configuring call for inspection point download – HYDRA-triggered

Inspection points from SAP are needed for all integration scenarios. The inspection point download is also executed by the hysapqmc.exe/out program. Use the [HYDRA Scheduler](#) to maintain entries for the download of inspection points.

Parameter name	Value
Product key	SAP-QMIDI
License key	SAP-QM-IDI
Command (Windows):	sh.exe .hysapqmc.scr /MESTYP_OUT=ZHYQMIDI_INSPPOINT /LOGSYS=<created logical system> /VARIANTE= ALL
Command (Unix):	./hysapqmc.scr /MESTYP_OUT=ZHYQMIDI_INSPPOINT /LOGSYS=<created logical system> /VARIANTE= ALL
Comment:	SAP-QMIDI: Download inspection points
Intervall	5

The HYDRA scope of delivery includes the “QM_IDI” variant for the QIRF_INSPPOINT_GETLIST function module as an example. The variant has mandatory to be copied into variant “ALL”.

To enable a dynamically operation, the parameter values for the parameters INSPLOT, INSPOPER, INSPPOINT_FROM and INSPPOINT_TO can be set to “USE_PROG_PARAMS”. In this case the program parameters of the same name will be evaluated for the function module call. I.e. :

**hysapqmc.exe/out /INSPLOT=000012345678 /INSPOPER=0010 /INSPPOINT_FROM=1
/INSPPOINT_TO=500**

If necessary multiple entries have to be created to download all necessary inspection points from SAP. It is strictly recommended to create the variants in the customer name space, starting with “U_”.

Configuring call for inspection point download – dynamically-triggered

Inspection points from SAP are needed for all integration scenarios. The inspection point download is also executed by the hysapqmc.exe/out program. It is executed based on downloaded inspection data. For configuration of the dynamically-triggered scenario several steps have to be performed:

- Configuration for the incoming QM-data message type:

The incoming QM data has to be processed twice – once to import the data into HYDRA and another time to download the inspection point. For that reason the script mle72imp_pp_qm.scr has to be used and edited accordingly.

Please note the importance to copy the script into the customer name space (starting “u_”) before changing it and calling it from the MLE distribution model.

- Additional parameters for the function module variant

In configuration table sap_fb_param_cfg additional parameters have to be inserted

Parameter name	Value
HY_SEGMENT_NAME	Segment from which inspection lot number and the operation number has to be taken from. DEFAULT: Z2QAIVC000X000
HY_SEGMENT_INSPLOT_FROM	Position in the specified segment where the inspection lot number starts
HY_SEGMENT_INSPLOT_TO	Position in the specified segment where the inspection lot number ends
HY_SEGMENT_INSOPERFROM	Position in the specified segment where the operation number starts
HY_SEGMENT_INSOPER_TO	Position in the specified segment where the operation number ends

- Programm parameter for hysapqmc.exe/out

The programm hysapqmc.exeout has to be executed with an additional programm parameter to enable dynamical download of inspection points:

/GET_INSPPOINTS_4_INSPLOTS



The functionality of dynamically downloading inspection points is available starting with hysapqmc.exe/out V8.1.1.26

Configuring for inspection point creation in HYDRA

When inspection points are not downloaded from SAP, they are created in HYDRA. The creation for time and quantity related inspection points is done by a cyclic process, that is scheduled in the HYDRA Scheduler. Use the [HYDRA Scheduler](#) to maintain an entry for the creation of inspection points.

Parameter name	Value
Product key	QMS-SQM
License key	QMS-SQM
Command (Windows):	sh.exe hyqmsipcr.scr
Command (Unix):	./hyqmsipcr.scr
Comment:	QMS-SQM: Creation of inspection points in HYDRA
Intervall	1

Configuration of confirmations

After data recording in HYDRA the confirmation data has to be transferred back to SAP. The confirmation is done on a cyclic base (Default: 15 minutes). Use the [HYDRA Scheduler](#) to maintain an entry for the creation of inspection points.

Parameter name	Value
Product key	SAP-QMIDI
License key	SAPQM-IDI
Command (Windows):	sh.exe ./qm_idi_rck.scr
Command (Unix):	./qm_idi_rck.scr
Comment:	SAP-QMIDI: Confirmation of inspection results HYDRA → SAP
Intervall	15



Confirming defect items to SAP it is possible that SAP only accepts single defect items. To transfer each defect item in a separate function call, the parameter `/SINGLE_IDOC` has to be added in `hysapupl.exe/out` command line for defect items inside the script `qm_idi_rck.scr`.

Configuring origin of results data

The origin of results data can be configured in SAP customizing. To enable HYDRA to transfer this information when uploading inspection results to SAP, the entry has to be customized in HYDRA accordingly. The customizing is only possible in HYDRA professional mode in the CAQ options:

Parameter name	Value
Option	1101
Option ID	0
Option Description	Origin of results data
Module	QMS
Value	<value as defined in SAP>
List	Yes

Configuring partial confirmations for inspection lots

Partial confirmations (transferring results before the final usage decision is made) for inspection lots can be configured HYDRA CAQ customizing.

Parameter name	Value
Option	1101
Option ID	0
Module	QMS
Value	<value as defined in SAP>
Addition	NONE [DIRECT]

By activating the CAQ option 1128 (value = Y) partial confirmations for inspection lots will be transferred to SAP even when the final usage decision is not done yet.

Remarks:

By activating the option WITHOUT the additional setting [DIRECT] the system will behave such as:

- Inspection points will be confirmed after closing them. Together with the inspection points single results and characteristic results will be transferred.
- For operations that are not inspection point relevant, the characteristic results with their single and sample results will be confirmed when closing the inspection order (usually when finishing the operation).
ATTENTION!! Setting the option might cause problems when reactivating CAQ-relevant operations.
- The usage decision will be confirmed when it is done, usually after closing the inspection requirement.

By activating the option WITH the additional setting [DIRECT] the system will behave such as:

- All inspection point details, recorded results und defect items will be confirmed directly after recording.
- The characteristics will be confirmed as closed, when the assigned inspection order is finished, usually when finishing the last operation.
ATTENTION!! Setting the option might cause problems when reactivating CAQ-relevant operations.
- Sample results will be confirmed as closed, when the assigned inspection order has been closed or the inspection point is closed.
ATTENTION!! Setting the option might cause problems when reactivating CAQ-relevant operations.
- The usage decision will be confirmed after recording, usually when the assigned inspection requirement is closed.



Activating the option with additional setting [DIRECT] is only available for testing reasons. The setting is NOT released for customers / productive usage.

In case the option is not available or inactive, all inspection lot data will be transferred after making the final usage decision.

Settings in HYDRA MES Link Enabling Inbound – Stand-alone CAQ

Use the [HYDRA distribution model](#) to maintain entries for HYDRA inbound processing:

Name of the parameter	Value
Download inspection specifications	
Message type	ZQM_IDI
Priority	None
Command	mle72imp.scr
Command parameter	/VARIANTE=<MLE variant to use>
Description	QM-IDI – Download inspection specifications
Log. Target system	Created logical system
Storage duration	10
Download catalogue master data	
Message type	ZHYQMIDI_CATALOG
Priority	None
Command	mle72imp.scr
Command parameter	/VARIANTE=<MLE variant to use>
Description	QM-IDI – Download catalogs
Log. Target system	Created logical system
Storage duration	10
Download inspection points	
Message type	ZHYQMIDI_INSPPOINT

Name of the parameter	Value
Priority	None
Command	mle72imp.scr
Command parameter	/VARIANTE=<MLE variant to use>
Description	
Log. Target system	Created logical system
Storage duration	10

Settings in HYDRA MES Link Enabling Inbound – PP-PDC integration

Use the [HYDRA distribution model](#) to maintain entries for HYDRA inbound processing for PP-PDC integration. Depending on the implementation sequence existing entries have to be changed and/or new entries have to be created.:

Name of the parameter	Value
Processing for PP-PDC message type PPCC2RECORDER:	
Message type	PPCC2RECORDER
Priority	None
Command	hysapqmc.scr
Command parameter	/VARIANTE =QM_IDI
Description	QM-IDI – Download inspection specifications
Log. Target system	Created logical system
Storage duration	10
Combined import of production order data and inspection specifications	
Message type	PPCC2RECORDER_QM_IDI

Name of the parameter	Value
Priority	None
Command	mle72imp.scr
Command parameter	/VARIANTE=<MLE variant to use>
Description	QM-IDI / PP-PDC: Import data
Log. Target system	Created logical system
Storage duration	10

Settings in HYDRA MES Link Enabling Inbound – SAP-ISS integration

Use the [HYDRA distribution model](#) to maintain entries for HYDRA inbound processing for PP-PDC and SAP-ISS integration. Depending on the implementation sequence existing entries have to be changed and/or new entries have to be created.:

Name of the parameter	Value
Processing for PP-PDC message type PPCC2RECORDER:	
Message type	PPCC2RECORDER
Priority	None
Command	hysapinf.scr
Command parameter	/REC_TYPE
Description	SAP-ISS – Download additional data
Log. Target system	Created logical system
Storage duration	10
Request inspection specifications:	
Message type	PPCC2HYINFOORDER

Name of the parameter	Value
Priority	None
Command	hysapqmc.scr
Command parameter	/VARIANTE =<MLE variant to use>
Description	QM-IDI – Request insp. specifications
Log. Target system	Created logical system
Storage duration	10
Combined import of production order data, additional data and inspection specifications	
Message type	PPCC2HYINFOORDER_QM_IDI
Priority	None
Command	mle72imp.scr
Command parameter	/VARIANTE=<MLE variant to use>
Description	QM / PP: Import data
Log. Target system	Created logical system
Storage duration	10

Settings in HYDRA MES Link Enabling outbound

Use the [HYDRA distribution model](#) to maintain entries for HYDRA outbound processing:

Name of the parameter	Value
To upload original values	
Message type	ZHYQMIDI_ORIGINAL_VALUES
Description	QM-IDI – Upload original values
IDoc-Typ	ZHYQMIDI_ORIGINAL_VALUES

Name of the parameter	Value
Storage duration	10
Log. Target system	<created logical system>
Segment name 1	Z2QAISE000X000
To upload sample values	
Message type	ZHYQMIDI_SAMPLE_VALUES
Description	QM-IDI – Upload sample values
IDoc-Typ	ZHYQMIDI_SAMPLE_VALUES
Storage duration	10
Log. Target system	<created logical system>
Segment name 1	Z2QAISR000X000
To upload feature values	
Message type	ZHYQMIDI_FEATURE_VALUES
Description	QM-IDI – Upload feature values
IDoc-Typ	ZHYQMIDI_FEATURE_VALUES
Storage duration	10
Log. Target system	<created logical system>
Segment name 1	Z2QAIMR000X000
To upload inspection points	
Message type	ZHYQMIDI_INSP_POINTS
Description	QM-IDI – Upload inspection point
IDoc-Typ	ZHYQMIDI_INSP_POINTS
Storage duration	10

Name of the parameter	Value
Log. Target system	<created logical system>
Segment name 1	Z2QAIPP000X000
To upload usage decisions	
Message type	ZHYQMIDI_USAGE_DECISION
Description	QM-IDI – Upload usage decision
IDoc-Typ	ZHYQMIDI_USAGE_DECISION
Storage duration	10
Log. Target system	<created logical system>
Segment name 1	Z2QAIVE000X000
To upload defect items	
Message type	ZHYQMIDI_DEFECT_ITEMS
Description	QM-IDI – Upload defect items
IDoc-Typ	ZHYQMIDI_DEFECT_ITEMS
Storage duration	10
Log. Target system	<created logical system>
Segment name 1	Z2QMIFE000X000